

Literature Review for TriGuard: Multimodal, Explainable and Locally Deployable Content Moderation

CM3070 Computer Science Final Project — CM3020 Artificial Intelligence Template, Project Idea
1: Orchestrating AI Models to Achieve a Goal

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1. Introduction

Online platforms now host text, images, audio and video side by side, and harmful content frequently spans more than one of these modalities at once (Kiela et al., 2020). Traditional automated moderators were designed to score a single signal — most commonly text — and struggle when a meme combines a benign caption with a hateful image, or when an innocuous post is wrapped around violent audio (Gorwa, Binns and Katzenbach, 2020). This survey reviews the work that informs **TriGuard**, a prototype CM3070 project that orchestrates three pre-trained perception models with a local Large Language Model (LLM) judge to produce a single, explainable moderation decision across text, image and audio.

The review covers eight sources spanning text moderation, multimodal fusion, audio recognition, the LLM-as-judge paradigm, vision–language pre-training, and the governance of algorithmic moderation. After describing the search strategy, each thematic section ends with a comparison paragraph that contrasts the work surveyed and points to the research gap that TriGuard addresses.

2. Search Strategy and Selection Criteria

Sources were retrieved from the University of London Online Library, Google Scholar, the ACM Digital Library, the NeurIPS and ICML proceedings, and the arXiv preprint server, using combinations of the terms *content moderation*, *toxicity detection*, *multimodal hate speech*, *hateful memes*, *audio event detection*, *LLM-as-judge*, *explainable AI* and *algorithmic moderation*. Inclusion criteria were: (i) the work is peer-reviewed or is a widely cited preprint with verifiable open weights, (ii) the work was published between 2017 and 2023 inclusive, and (iii) the work either contributes a method TriGuard plans to use, contributes a benchmark TriGuard plans to evaluate against, or critically frames the governance context. Blog posts and product pages were excluded from the primary citations and used only as supporting documentation.

3. Text-Based Moderation

The strongest production exemplar of text moderation is Google Jigsaw’s **Perspective API**. Lees et al. (2022) replace the API’s earlier per-language BERT classifiers with a single multilingual character-level transformer and report parity or improvement on internal toxicity benchmarks while serving billions of daily requests. Their account is significant for two reasons. First, it demonstrates that a single model can serve many languages — a property essential for any moderation system that hopes to be useful outside English. Second, it formalises the operational realities (latency budgets, fairness audits, drift monitoring) that academic moderation papers tend to ignore.

The limitation is structural rather than incidental. Perspective scores plain text. It cannot read a meme, cannot transcribe a voice note, and ships no human-readable rationale beyond a numeric probability per attribute. The

system is closed-weights and cloud-only, which means platforms with privacy-sensitive content (medical, legal, child safeguarding) cannot deploy it without exposing user data to a third party. Compared to the multimodal and explainable work reviewed below, Perspective is best understood as a strong but deliberately narrow baseline — one that any new moderation pipeline must justify itself against on the text track, but that cannot, in isolation, satisfy the multimodal harms documented in §4–§5.

4. Image-Text and Hateful Meme Moderation

The single most influential resource for multimodal moderation research is the **Hateful Memes Challenge** introduced by Kiela et al. (2020). The benchmark contains roughly 10 000 memes constructed so that unimodal text classifiers and unimodal image classifiers score close to chance, forcing models to fuse modalities. Human accuracy is reported at approximately 85 %, while strong unimodal baselines reach only ~ 50 %. The first generation of multimodal baselines — VisualBERT (Li, L.H. et al., 2019) and ViLBERT — push performance into the mid-60s but remain substantially below humans, and the authors flag that synthetic “confounder” memes (where flipping a single token changes the label) defeat shortcuts that unimodal models rely on.

VisualBERT itself remains the canonical fusion baseline. Li, L.H. et al. (2019) propose a single-stream transformer that concatenates word embeddings and Faster-RCNN region features. The architecture is simple and reproducible — virtues that have made it a default starting point — but it is also brittle: caption quality depends on the region detector, and the only output is a class score with no rationale. **BLIP** (Li, J. et al., 2022) addresses the brittleness by removing the region-extraction stage and unifying captioning, retrieval and visual question answering inside one vision–language model with open weights. BLIP captions are short natural-language descriptions of an image, which is precisely the form an LLM judge can reason about.

Comparing across these three works exposes a recurring pattern. The benchmark literature (Kiela et al., 2020) shows that fusion *helps*; the architecture literature (Li, L.H. et al., 2019; Li, J. et al., 2022) shows that fusion is *feasible* with open components; yet none of these works delivers an end-to-end moderation system, and none produces an output that a human moderator could audit. Audio is absent from all three.

5. Audio and Speech-Based Moderation

Audio moderation is the least developed of the three modalities in the literature surveyed. **AudioSet** (Gemmeke et al., 2017) supplies the dominant ontology and benchmark for general-purpose audio event recognition — over two million ten-second clips across 632 sound event classes — and the MobileNet/CNN baselines released alongside it, of which YAMNet is the most widely re-used, provide pretrained features that work on a CPU. AudioSet is broad rather than safety-focused: it contains classes such as “shout”, “explosion”, “weapon”, “screaming” that are relevant for moderation, but no fine-grained taxonomy of harm.

The complement to YAMNet on the speech side is **Whisper** (Radford et al., 2023), an encoder–decoder transformer trained on 680 000 hours of weakly supervised web audio. Whisper reports near-human English ASR and viable multilingual transcription across 96 languages, all with open weights that run offline. Crucially for TriGuard, Whisper produces *text*, which can then be fed into the text moderation track described in §3 — a clean composition that avoids inventing a new “audio toxicity” classifier from scratch.

Comparing the two: AudioSet/YAMNet gives a coarse “what’s happening?” tag (shout, glass breaking, music), while Whisper gives the verbatim words said. Neither, on its own, decides whether the audio is harmful; both are evidence to be combined. The existing literature does not appear to have packaged speech-to-text and audio-event classification together for moderation, which is part of the gap TriGuard targets.

6. Model Orchestration and LLM-as-Judge Approaches

The orchestration layer that combines the three perception models draws on the **LLM-as-a-judge** literature. Zheng et al. (2023) systematically evaluate strong LLMs (GPT-4 in particular) as judges of open-ended responses on MT-Bench and Chatbot Arena. Their headline finding — that LLM judges agree with crowd-worker

preferences roughly 80 % of the time — is matched by an equally important set of cautionary findings: judges exhibit *position bias* (preferring the first option presented), *verbosity bias* (preferring longer answers) and *self-preference bias* (preferring outputs from the same model family). Crucially, the authors show that these biases can be partially mitigated with calibration, swapped-order prompting and structured rubrics.

TriGuard adapts this pattern to a different setting. The judge does not compare two candidate answers; it reads a structured *evidence packet* (text toxicity scores, image caption, audio transcript + event tags) and returns a risk score, a risk label, the flagged modalities and a written rationale. Position and length bias are less acute when the input is fixed-format, but hallucinated reasoning — claims unsupported by the evidence — remains a live risk and motivates the prompt-engineering and JSON-schema enforcement TriGuard’s design adopts.

Compared with the fusion-network approach of §4, the LLM-judge approach trades some accuracy headroom for a major qualitative gain: the decision is *expressible in natural language*. That gain becomes the bridge to the explainability and governance literature.

7. Explainability, Privacy, and Responsible Moderation

The technical literature reviewed so far treats moderation as a classification problem. **Gorwa, Binns and Katzenbach (2020)** reframe it as a sociotechnical one. Their survey distinguishes hash-matching, classifier-based, and policy-driven moderation, and argues that opaque automated decisions create governance failures — users denied due process, regulators unable to audit, and platforms unable to explain themselves — that purely technical fixes do not solve. The paper predates the EU Digital Services Act and the UK Online Safety Act, but the transparency requirements those instruments now codify (Statement of Reasons under DSA Article 17; risk-assessment duties under Article 34) closely track the concerns Gorwa and colleagues raise.

Read together with the technical works above, this paper sharpens the brief. A moderation pipeline that scores well on Hateful Memes but cannot explain itself fails the governance test. A pipeline that explains itself but routes content through a closed cloud API fails the privacy test. A pipeline that is local and explainable but ignores audio fails the completeness test. The contribution of Gorwa, Binns and Katzenbach (2020) to TriGuard’s design is to make these three failure modes legible as *requirements* rather than as nice-to-haves.

In comparison with §3–§6, this is the only source that frames moderation as accountable to people who do not write code. That framing supplies the justification for TriGuard’s two non-technical design commitments: a human-readable rationale on every decision, and local-only deployment by default.

8. Comparative Analysis Table

The eight sources span complementary axes: modality coverage, output explainability, deployability and governance framing. The matrix below condenses the comparison; the full ten-column matrix is reproduced in the project repository at `docs/literature_matrix.md`.

Source	Modality covered	Output type	Open weights	Explanation?
Lees et al. (2022)	Text	Class scores	No	No
Kiela et al. (2020)	Image + text	Benchmark labels	Yes	No
Li, L.H. et al. (2019)	Image + text	Class score	Yes	No
Gemmeke et al. (2017)	Audio	Event tags	Yes	No
Radford et al. (2023)	Audio (speech)	Text transcript	Yes	No
Zheng et al. (2023)	n/a (orchestration)	Judgement + rationale	Mixed	Yes
Li, J. et al. (2022)	Image + text	Caption	Yes	Indirectly
Gorwa et al. (2020)	n/a (governance)	n/a	n/a	Argues <i>for</i> explanation

9. Research Gap

Synthesising across the matrix, the gap is concrete:

Existing moderation systems often focus on one modality or rely on closed cloud-based APIs. Prior multimodal work shows the value of combining text and image, but many systems do not include audio, do not run locally, and do not provide human-readable explanations. TriGuard addresses this gap by designing a local orchestration pipeline that combines text, image, and audio evidence and produces an explainable decision for human review.

No single source surveyed delivers all four properties — **multimodal, audio-aware, explainable, locally deployable** — at once. TriGuard sits at their intersection.

10. How the Literature Informs TriGuard

Each source contributes a concrete design implication.

- **Lees et al. (2022)** justifies using a strong off-the-shelf text classifier for the text wrapper rather than training a new one; sets the text-track baseline.
- **Kiela et al. (2020)** supplies the evaluation benchmark for the image-text track and quantifies the gap between unimodal and fusion approaches that motivates orchestration.
- **Li, L.H. et al. (2019)** acts as the academic baseline against which TriGuard’s “LLM judge over evidence summaries” must be contrasted in the final evaluation.
- **Gemmeke et al. (2017)** supplies the YAMNet pretrained weights and event ontology for the audio-event wrapper.
- **Radford et al. (2023)** supplies the Whisper speech-to-text model that allows the text wrapper to be reused on audio content.
- **Zheng et al. (2023)** validates the LLM-as-judge pattern, identifies the biases TriGuard’s prompting and schema must mitigate, and motivates the structured-JSON output contract.
- **Li, J. et al. (2022)** supplies BLIP captions, which are the input format the LLM judge needs from the image side.
- **Gorwa, Binns and Katzenbach (2020)** anchors the explainability and local-deployment requirements that distinguish TriGuard from a purely accuracy-driven research prototype.

Together, the eight works define both the methodological scaffold and the policy frame of the project. The project design document and the evaluation protocol that follow this review map each of these implications to a concrete component and a concrete evaluation track, so that the gap statement above is testable rather than rhetorical.

11. References

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